

```
PS C:\Users\LAB 102 PC 10\Desktop\s241210108\lab5> pyth
Path: ['A', 'B', 'D', 'G']
Moves: 3
PS C:\Users\LAB 102 PC 10\Desktop\s241210108\lab5> 
```

(5

```
PS C:\Users\LAB 102 PC 10>
You are O, AI is X
1 | 2 | 3
-----
4 | 5 | 6
-----
7 | 8 | 9

Enter position (1-9): 2
1 | 0 | 3
-----
4 | 5 | 6
-----
7 | 8 | 9

AI move:
X | 0 | 3
-----
4 | 5 | 6
-----
7 | 8 | 9

Enter position (1-9): 6
X | 0 | 3
-----
4 | 5 | 0
-----
7 | 8 | 9

Enter position (1-9): 4
X | 0 | 3
-----
0 | 5 | 0
-----
X | 8 | 9

AI move:
X | 0 | 3
-----
0 | X | 0
-----
X | 0 | 9

AI move:
X | 0 | X
-----
0 | X | 0
-----
X | 0 | 9

AI wins!
```

```

PS C:\Users\LAB 102 PC 10\Desktop\s241210108\lab5> python hillClimbing.py
Prev step = 9.74371404014271 Value = 5.060036703925846
Prev step = 9.487428080285419 Value = 9.98870842141173
Prev step = 9.231142120428128 Value = 14.78601515245768
Prev step = 8.974856160570837 Value = 19.45195689706368
Prev step = 8.718570200713547 Value = 23.98653365522975
Prev step = 8.462284240856256 Value = 28.389745426955855
Prev step = 8.205998280998966 Value = 32.66159221224203
Prev step = 7.949712321141676 Value = 36.80207401108823
Prev step = 7.693426361284386 Value = 40.81119082349449
Prev step = 7.437140401427096 Value = 44.688942649460806
Prev step = 7.180854441569807 Value = 48.43532948898718
Prev step = 6.924568481712517 Value = 52.05035134207361
Prev step = 6.668282521855227 Value = 55.534008208720095
Prev step = 6.411996561997937 Value = 58.88630008892663
Prev step = 6.155710602140648 Value = 62.10722698269323
Prev step = 5.899424642283358 Value = 65.19678889001987
Prev step = 5.643138682426068 Value = 68.15498581090658
Prev step = 5.386852722568778 Value = 70.98181774535334
Prev step = 5.130566762711489 Value = 73.67728469336015
Prev step = 4.874280802854199 Value = 76.24138665492703
Prev step = 4.617994842996909 Value = 78.67412363005396
Prev step = 4.361708883139619 Value = 80.97549561874094
Prev step = 4.10542292328233 Value = 83.14550262098797
Prev step = 3.84913696342504 Value = 85.18414463679507
Prev step = 3.59285100356775 Value = 87.0914216661622
Prev step = 3.3365650437104604 Value = 88.86733370908941
Prev step = 3.0802790838531706 Value = 90.51188076557668
Prev step = 2.823993123995881 Value = 92.02506283562398
Prev step = 2.567707164138591 Value = 93.40687991923136
Prev step = 2.3114212042813014 Value = 94.65733201639878
Prev step = 2.0551352444240116 Value = 95.77641912712626
Prev step = 1.7988492845667219 Value = 96.76414125141379
Prev step = 1.5425633247094321 Value = 97.62049838926139
Prev step = 1.2862773648521424 Value = 98.34549054066903
Prev step = 1.0299914049948526 Value = 98.93911770563673
Prev step = 0.7737054451375628 Value = 99.40137988416448
Prev step = 0.5174194852802729 Value = 99.7322770762523
Prev step = 0.26113352542298307 Value = 99.93180928190016
Prev step = 0.004847565565693213 Value = 99.99997650110808

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PS C:\Users\win 11\Desktop\4th Sem\AI Codes> python -u "c:\U
Solution found:
{'A': 4, 'L': 5, 'S': 8, 'G': 1, 'E': 3, 'B': 7, 'M': 9}
7483 + 7455 = 14938
PS C:\Users\win 11\Desktop\4th Sem\AI Codes>

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```

PS C:\Users\Win 11\D . R . # #
PS C:\Users\Win 11\D . # . . #
===== MINIMAX =====
STATE:
A . . . #
. # . . R
. R . # #
. # . . #
R . . B R

Player A (MAX)

STATE:
. . . . #
A # . . R
. R . # #
. # . . #
R . . B R

Player B (MIN)

STATE:
. . . . #
A # . . R
. R . # #
. # . B #
R . . . R

Utility: -4

STATE:
. . . . #
A # . . R
. R . # #
. # . . #
. R . # #
. # . . R
R . . . B

Utility: -4
Decision (B): -4
Decision (A): -4

===== ALPHA-BETA =====
Visiting Node:
A . . . #
. # . . R
. R . # #
. # . . #
R . . B R

Visiting Node:
. . . . #
A # . . R
. R . # #
. # . . #
R . . B R

Visiting Node:
. . . . #
A # . . R
. R . # #
. # . B #
R . . . R

Utility: -4
PRUNED (B)

===== RESULTS =====
Minimax Value: -4
Alpha-Beta Value: -4
Nodes Visited: 7
Branches Pruned: 1
Winner: B
PS C:\Users\Win 11\Des

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Utility: -4

STATE:
. . . . #
A # . . R
. R . # #
. # . . #
R . . . B

Utility: -3
Decision (B): -4

STATE:
. A . . #
. # . . R
. R . # #
. # . . #
R . . B R

Player B (MIN)

STATE:
. A . . #
. # . . R
. R . # #
. # . B #
R . . . R

Utility: -4

STATE:
. A . . #
. # . . R
. R . # #
. # . . #

```

```

nitdelhi@pc:~$ vi family.pl
nitdelhi@pc:~$ swipl
Welcome to SWI-Prolog (threaded, 64 bits, version 7.6.4)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit http://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- [family].
true.

?- father(john,mary).
true.

?- mother(mary,susan).
true.

?- parent(X,mary).
X = john.

```

```

nitdelhi@pc:~$ vi numbers.pl
nitdelhi@pc:~$ swipl
Welcome to SWI-Prolog (threaded, 64 bits, version 7.6.4)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit http://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- [numbers].
ERROR: /home/nitdelhi/numbers.pl:1:
    No permission to modify static procedure 'number/1'
ERROR: /home/nitdelhi/numbers.pl:2:
    No permission to modify static procedure 'number/1'
ERROR: /home/nitdelhi/numbers.pl:3:
    No permission to modify static procedure 'number/1'
ERROR: /home/nitdelhi/numbers.pl:4:
    No permission to modify static procedure 'number/1'
true.

?- even(4).
true.

?- even(3).
false.

?- odd(5).
true.

?- odd(4).
false.

```

```

nitdelhi@pc:~$ vi hello.pl
nitdelhi@pc:~$ [hello].
[hello].: command not found
nitdelhi@pc:~$ swipl
Welcome to SWI-Prolog (threaded, 64 bits, version 7.6.4)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit http://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- [hello].
true.

?- main.
Hello World
true.

```



```

PS C:\Users\LAB 102 PC 25\Desktop\241210092AI> python -u "c:\Users\LAB 102 PC 25\Desktop\241210092AI\Goal_reached.py"
Goal reached!

Path taken: [(0, 0), (1, 0), (1, 1), (2, 1), (2, 2), (2, 3), (3, 3), (4, 3), (4, 4)]

Grid with path:
S 0 1 0 0
* * 1 0 1
1 * * * 0
1 1 1 * 1
0 0 0 * G
PS C:\Users\LAB 102 PC 25\Desktop\241210092AI>

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PS C:\Users\LAB 102 PC 25\Desktop\241210092AI> python -u "c:\Users\LAB 102 PC 25\Desktop\241210092AI\Failed_to_find_solution.py"
Failed to find solution (Random Walk)
Random Walk Board:
. . Q . . Q . .
. . . . Q . . .
. . . Q . . . .
. Q . . . . . .
Q . . . . . . .
. . . . . . . .
. . . . . . Q Q
. . . . . . . .

Conflicts: 7
Stopped (Simulated Annealing)
Simulated Annealing Board:
. Q . . . . . .
. . . . Q . . .
. . . . . . . Q
Q . . . . . . .
. . . . . Q . .
. . . Q . . . .
. . . . Q . . .
. . Q . . . . .

Conflicts: 1
PS C:\Users\LAB 102 PC 25\Desktop\241210092AI>

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PS C:\Users\LAB 102 PC 25\Desktop\241210092AI> python -u "c:\Users\LAB 102 PC 25\Desktop\241210092AI\Solved_in_generation.py"
Solved in generation: 24
Solution: [2, 5, 1, 4, 7, 0, 6, 3]
PS C:\Users\LAB 102 PC 25\Desktop\241210092AI>

```

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